

CISCO PACKET TRACER · CCNA ENTERPRISE SERIES

CCNA Enterprise Lab Roadmap

One-Day Progressive Lab Guide

- % 11 Progressive Labs · Simple !' Enterprise Grade
- % Full Topology Diagrams · Cabling Tables · IP Plans
- % Error-Proof Commands with Line-by-Line Explanations
 - % Every Common Error + Exact Fix Included
 - % Verification Commands with Expected Output
- % Capstone: Real-World Multi-Site Enterprise Network

Technologies Covered:

VLANs · Trunking · STP · OSPF · HSRP · DHCP · NAT/PAT · ACLs · SSH · Port Security · DHCP Snooping · DAI

Packet Tracer 8.x Compatible · CCNA 200-301 Aligned · Enterprise Portfolio Ready

Build. Verify. Master.

Start from zero. End with an enterprise network running on your machine today.

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HOW TO USE THIS GUIDE

This guide takes you from zero to an enterprise-grade network in one day. Each lab builds directly on the previous one — skills you learn in Lab 1 are used in Lab 11. Follow in order.

REQUIRED TOOLS

- Cisco Packet Tracer 8.2 or later — free download at netacad.com (create a free account)
- This PDF — open it on a second monitor or print it
- Notepad/text editor — to paste configs before applying them
- At least 4 GB RAM free for Packet Tracer with large topologies

SUGGESTED DAILY SCHEDULE

Time	Labs	Focus
8:00 – 9:30 AM	Lab 1 + Lab 2	Foundation — Switches, VLANs
9:30 – 11:00 AM	Lab 3 + Lab 4	Routing — Static + Inter-VLAN
11:00 AM – 12:30 PM	Lab 5 + Lab 6	STP + OSPF
12:30 – 1:00 PM	BREAK	Rest, review notes
1:00 – 2:30 PM	Lab 7 + Lab 8	DHCP + NAT/PAT
2:30 – 4:00 PM	Lab 9 + Lab 10	ACLs + HSRP
4:00 – 6:30 PM	Lab 11	Enterprise Capstone
6:30 – 7:00 PM	Review	Screenshot results for LinkedIn

TIP

Each lab has a SAVE CHECKPOINT at the end. Before starting the next lab, save your Packet Tracer file with the lab number (e.g. "Lab01-Complete.pkt"). This way you can always go back if something breaks.

WARNING

If a command gives an "Invalid input detected" error, you are either on the wrong device, in the wrong mode, or there is a typo. Check the mode prompt:

- Router> means USER EXEC mode (very limited)
- Router# means PRIVILEGED EXEC mode (show commands work here)
- Router(config)# means GLOBAL CONFIG mode (where you configure things)
- Router(config-if)# means INTERFACE CONFIG mode

OBJECTIVES

- Configure hostnames, login banners, and MOTD on routers and switches
- Set encrypted enable secret and console/VTY passwords
- Enable SSH v2 and disable Telnet for secure remote management
- Disable DNS lookup to prevent command typo delays
- Encrypt all plaintext passwords in the running config

CORE CONCEPTS — READ FIRST

Before typing a single command, understand what you are securing and why:

ENABLE SECRET vs ENABLE PASSWORD: Always use "enable secret" — it stores the password as an MD5 hash. "enable password" stores it in plaintext or weak reversible encryption. If both exist, enable secret takes priority. You will ONLY use enable secret in this course.

PRIVILEGE LEVELS: Cisco IOS has 16 privilege levels (0-15). Level 1 = user exec (very limited). Level 15 = full access. When you type "enable" and authenticate, you jump to level 15. The enable secret protects this jump.

service password-encryption: This command encrypts all CURRENT and FUTURE plaintext passwords in the config using Cisco type-7 (reversible) encryption. It is weak but better than plaintext. The enable secret is already hashed (type-5 MD5) regardless of this command.

SSH vs TELNET: Telnet sends everything — including passwords — as plaintext over the network. Anyone running a packet capture can see your credentials. SSH encrypts the entire session. In 2024, Telnet is never acceptable on any production device.

RSA Key Size: SSH requires an RSA key pair. Cisco recommends 2048-bit minimum. The command "crypto key generate rsa modulus 2048" generates this. After generation, the device can accept SSH connections. You MUST set a hostname and domain-name FIRST or the key generation will fail — Cisco uses "hostname.domain-name" as the key identifier.

TOPOLOGY

R1 — Global Configuration

! % % % FIRST: Enter privileged exec mode % % %
enable

! (No password yet — just press Enter)

! % % % Enter global config mode % % %
configure terminal

! Shortcut: "conf t" works too

! % % % Set hostname % % %
hostname R1

! Prompt changes to: R1(config)#

! % % % Set domain name (REQUIRED for SSH) % % %
ip domain-name novatech.local

! % % % Disable DNS lookup (prevents hanging on typos) % % %
no ip domain-lookup

! Without this, a mistyped command like "shw" causes
! the router to try DNS lookup and hang for 30 seconds

! % % % Encrypt all plaintext passwords % % %
service password-encryption

! % % % Set the privileged exec password (ALWAYS use secret) % % %
enable secret NovaTech@2024!

! % % % Create local user for SSH login % % %
username admin privilege 15 secret Admin@NovaTech2024

! % % % Generate RSA key for SSH (hostname + domain MUST be set first) % % %
crypto key generate rsa modulus 2048

! When prompted "How many bits in the modulus [512]:" ! type 2048 Enter

! Output: "% Generating 2048 bit RSA keys, keys will be non-exportable..."

! % % % Enable SSH version 2 % % %

ip ssh version 2

ip ssh time-out 60

ip ssh authentication-retries 3

R1 — Line Configuration

```
! % % % Secure Console port % % %
line console 0
  password C0ns0le@NovaT
  login
  logging synchronous
! logging synchronous prevents log messages from interrupting your typing
  exec-timeout 5 0
! exec-timeout 5 0 = auto-logout after 5 minutes of inactivity
  exit

! % % % Secure VTY lines (Telnet/SSH remote access) % % %
line vty 0 4
  login local
! "login local" = use the local user database (admin/Admin@NovaTech2024)
  transport input ssh
! "transport input ssh" = ONLY allow SSH, reject Telnet attempts
  exec-timeout 10 0
  exit
```

R1 — Interface + Banner + Save

```
! % % % MOTD Banner (shown BEFORE login) % % %
banner motd ^
=====
  AUTHORIZED ACCESS ONLY — NovaTech Solutions
  Unauthorized access is PROHIBITED and will be prosecuted.
  All sessions are monitored and logged.
=====
^
! The ^ is a delimiter — everything between the two ^ is the banner
! You can use any character as delimiter as long as it does not appear in the text

! % % % Configure interface % % %
interface GigabitEthernet0/0
  ip address 192.168.1.1 255.255.255.0
  description *** LINK TO SW1 ***
  no shutdown
! "no shutdown" activates the interface — router interfaces are SHUTDOWN by default
  exit

! % % % Save configuration % % %
end
copy running-config startup-config
! OR shortcut: "wr" (write memory)
! Output: "Destination filename [startup-config]?" ! press Enter
```

2 Configure Switch SW1 — Click SW1 !' CLI tab

SW1 — Full Configuration

```
enable
configure terminal

hostname SW1
ip domain-name novatech.local
no ip domain-lookup
service password-encryption

enable secret NovaTech@2024!
username admin privilege 15 secret Admin@NovaTech2024

crypto key generate rsa modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3

! % % % Management IP on VLAN 1 (default VLAN) % % %
interface Vlan1
  ip address 192.168.1.2 255.255.255.0
  no shutdown
  exit

! % % % Default gateway so switch can reach other networks % % %
ip default-gateway 192.168.1.1

line console 0
  password C0ns0le@NovaT
  login
  logging synchronous
  exec-timeout 5 0
  exit

line vty 0 15
! Switches use vty 0 15 (16 lines) vs routers vty 0 4 (5 lines)
  login local
  transport input ssh
  exec-timeout 10 0
  exit

banner motd ^
=====
  AUTHORIZED ACCESS ONLY — SW1 NovaTech Solutions
=====
^

end
copy running-config startup-config
```

3 Set PC1 IP Address

- Click PC1 !' Desktop tab !' IP Configuration
- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
-

Click close

4 Verify Lab 1

Verification Commands

```
! % % On R1 % %  
show running-config  
! Look for: "enable secret 5 $1$..." (the 5 means MD5 hash)  
! Look for: "password 7 ..." (the 7 means type-7 encrypted by service password-encryption)  
  
show ip ssh  
! Expected: SSH Enabled - version 2.0  
  
show users  
! Shows who is currently logged in  
  
! % % From PC1 Desktop ! Command Prompt % %  
ping 192.168.1.1  
! Expected: Reply from 192.168.1.1 ... (4 replies)  
  
ping 192.168.1.2  
! Expected: Reply from 192.168.1.2 ... (4 replies)  
  
! % % Try SSH from PC1 ! Desktop ! SSH % %  
! Host: 192.168.1.1, Username: admin, Password: Admin@NovaTech2024
```

COMMON ERROR

If "show ip ssh" says "SSH disabled", it means either: (1) You did not set a hostname + domain-name before generating the RSA key, or (2) You did not run "ip ssh version 2". Fix: Re-enter hostname, domain-name, then "crypto key zeroize rsa" then "crypto key generate rsa modulus 2048".

OBJECTIVES

- Create VLANs and assign descriptive names in the VLAN database
- Assign switch ports to specific VLANs as access ports
- Configure 802.1Q trunk links between switches
- Set the native VLAN to a non-default value (security hardening)
- Verify VLAN and trunk configuration with show commands

CORE CONCEPTS — READ FIRST

WHY VLANs? Without VLANs, every device connected to a switch is in the same broadcast domain. A single broadcast (ARP request, DHCP discover) floods to ALL ports. VLANs create separate, isolated broadcast domains on a single physical switch — departments cannot talk to each other without going through a router, even if they share the same physical hardware.

ACCESS PORT vs TRUNK PORT: An access port belongs to exactly ONE VLAN. Traffic in/out of an access port is UNTAGGED — the connected device (PC, printer, phone) has no idea VLANs exist. A trunk port carries traffic for MULTIPLE VLANs simultaneously. Each frame on a trunk port is tagged with a 4-byte 802.1Q header containing the VLAN ID. Trunk ports connect switches to switches, or switches to routers.

NATIVE VLAN: One VLAN on each trunk is designated as "native." Frames in the native VLAN are sent UNTAGGED on the trunk (no 802.1Q header). The default native VLAN is VLAN 1 — this is a well-known security risk called "VLAN hopping." Always change the native VLAN to an unused VLAN (we use VLAN 99). BOTH ends of a trunk MUST agree on the native VLAN or you get a CDP native VLAN mismatch warning and traffic corruption.

DTP (Dynamic Trunking Protocol): Cisco proprietary protocol that auto-negotiates trunk links. It is a security risk because an attacker can send DTP frames to force a port into trunk mode and access all VLANs. ALWAYS disable DTP by using "switchport mode access" on access ports and "switchport nonegotiate" on trunk ports.

TOPOLOGY

[illegible]

Device	Interface	Connected To	Cable Type	Mode
SW1	Fa0/1	PC-DATA-1	Straight-Through	Access VLAN 10
SW1	Fa0/11	PC-VOICE-1	Straight-Through	Access VLAN 20
SW1	Gi0/1	SW2 Gi0/1	Crossover / Auto	Trunk (native 99)
SW2	Fa0/1	PC-DATA-2	Straight-Through	Access VLAN 10
SW2	Fa0/11	PC-VOICE-2	Straight-Through	Access VLAN 20
SW2	Gi0/1	SW1 Gi0/1	Crossover / Auto	Trunk (native 99)

1 Configure SW1 — VLANs + Access Ports + Trunk

SW1 — VLANs and Access Ports

```
enable
configure terminal

! % % % STEP 1: Create VLANs in the VLAN database % % %
vlan 10
  name DATA
vlan 20
  name VOICE
vlan 30
  name MANAGEMENT
vlan 99
  name NATIVE
exit

! Note: VLANs are only valid when created here AND when the port is assigned.
! If you assign a port to VLAN 10 before creating VLAN 10, Packet Tracer
! creates it automatically — but always create VLANs explicitly in production.

! % % % STEP 2: Configure access ports for DATA (VLAN 10) % % %
interface range FastEthernet0/1 - 10
  switchport mode access
! Forces the port into access mode — disables DTP negotiation
  switchport access vlan 10
! Assigns this port to VLAN 10
  switchport nonegotiate
! Disables DTP — port will NOT send DTP frames
  spanning-tree portfast
! PortFast: skips STP listening/learning states
! ONLY use on ports connected to end devices, NEVER on trunk ports!
  spanning-tree bpduguard enable
! BPDUGuard: if a BPDU is received (switch connected), port goes into err-disable
! This prevents someone plugging in a switch on a user port
  no shutdown
exit

! % % % STEP 3: Configure access ports for VOICE (VLAN 20) % % %
interface range FastEthernet0/11 - 20
  switchport mode access
  switchport access vlan 20
  switchport nonegotiate
  spanning-tree portfast
  spanning-tree bpduguard enable
  no shutdown
exit
```

SW1 — Trunk Port + Management

```
! % % % STEP 4: Configure trunk port to SW2 % % %
interface GigabitEthernet0/1
  switchport mode trunk
! Forces trunk mode — does NOT wait for DTP negotiation
  switchport trunk encapsulation dot1q
! Specifies 802.1Q encapsulation (some switches need this, some do not)
! If you get "Command rejected" on this line, your switch auto-uses dot1q — skip it
  switchport trunk native vlan 99
! Changes native VLAN from 1 to 99 (security hardening)
! WARNING: BOTH ENDS must use the same native VLAN or traffic will be misassigned
  switchport trunk allowed vlan 10,20,30,99
! Only allows specified VLANs on this trunk — "pruning" unused VLANs
! This reduces unnecessary broadcast traffic
  switchport nonegotiate
! Disable DTP on trunk port too
  no shutdown
exit

! % % % STEP 5: Management IP on VLAN 30 % % %
interface Vlan30
  ip address 192.168.30.2 255.255.255.0
  no shutdown
exit

ip default-gateway 192.168.30.1

end
copy running-config startup-config
```

Apply the EXACT same configuration to SW2 (change the management IP to 192.168.30.3). The trunk port on SW2 Gi0/1 must have the SAME native VLAN (99) and allowed VLANs as SW1.

2 Set PC IP Addresses

PC	IP Address	Subnet Mask	Gateway
PC-DATA-1	192.168.10.10	255.255.255.0	192.168.10.1 (not set yet — Lab 4)
PC-DATA-2	192.168.10.11	255.255.255.0	192.168.10.1
PC-VOICE-1	192.168.20.10	255.255.255.0	192.168.20.1
PC-VOICE-2	192.168.20.11	255.255.255.0	192.168.20.1

3 Verify VLANs and Trunking

Verification

! On SW1 — verify VLAN database

show vlan brief

! Expected output:

! VLAN Name	Status	Ports
! 1 default	active	
! 10 DATA	active	Fa0/1, Fa0/2 ... Fa0/10
! 20 VOICE	active	Fa0/11 ... Fa0/20
! 30 MANAGEMENT	active	
! 99 NATIVE	active	

! Verify trunk links

show interfaces trunk

! Expected output:

! Port	Mode	Encapsulation	Status	Native vlan
! Gi0/1	on	802.1q	trunking	99

! Port Vans allowed on trunk

! Gi0/1 10,20,30,99

! Verify specific interface mode

show interfaces FastEthernet0/1 switchport

! Look for:

! Administrative Mode: static access

! Operational Mode: static access

! Access Mode VLAN: 10

! Connectivity test: PC-DATA-1 ! PC-DATA-2 (same VLAN, different switch)

! From PC-DATA-1 cmd: ping 192.168.10.11

! Expected: SUCCESS (they are in the same VLAN, trunk carries VLAN 10)

! This should FAIL (different VLANs, no router yet):

! From PC-DATA-1 cmd: ping 192.168.20.10

! Expected: Request timeout (no inter-VLAN routing yet)

COMMON ERROR

COMMON ERROR: "Native VLAN mismatch" warning in SW1 console
CAUSE: SW1 native VLAN is 99 but SW2 native VLAN is still 1 (default)
FIX: Configure "switchport trunk native vlan 99" on SW2 Gi0/1 as well
Both ends of a trunk MUST have the same native VLAN.

- Configure IP addresses on router interfaces
- Add static routes to reach remote networks
- Configure a default static route (gateway of last resort)
- Verify routing table and test end-to-end connectivity

DEFAULT ROUTE: The "route of last resort" — used when no specific route matches the destination. Configured as "ip route 0.0.0.0 0.0.0.0 [next-hop]". Every packet destined for an unknown network gets forwarded here. Typically points toward the ISP or internet gateway.

[illegible]

Device	Interface	IP Address	Connected To
R1	Gi0/0	10.1.1.1/24	PC-A (via switch or direct)
R1	Gi0/1	10.0.0.1/30	R2 Gi0/1 — WAN link
R2	Gi0/0	10.2.2.1/24	PC-B (via switch or direct)
R2	Gi0/1	10.0.0.2/30	R1 Gi0/1 — WAN link
PC-A	Fa0	10.1.1.10/24	R1 Gi0/0 (or SW Fa0/x)

Fa0

10.2.2.10/24

R2 Gi0/0 (or SW Fa0/x)

The /30 mask (255.255.255.252) gives you exactly 2 usable IPs — perfect for point-to-point router links. You never waste addresses. 10.0.0.0/30 gives: .1 and .2 as usable IPs, .0 as network, .3 as broadcast.

1 Configure Router R1

R1 — Static Route Configuration

```
enable
configure terminal
hostname R1
no ip domain-lookup

! % % % LAN interface (toward PC-A) % % %
interface GigabitEthernet0/0
description *** LAN — toward PC-A ***
ip address 10.1.1.1 255.255.255.0
no shutdown
exit

! % % % WAN interface (toward R2) % % %
interface GigabitEthernet0/1
description *** WAN — toward R2 ***
ip address 10.0.0.1 255.255.255.252
no shutdown
exit

! % % % Static route to R2's LAN network % % %
ip route 10.2.2.0 255.255.255.0 10.0.0.2
! Read as: "To reach 10.2.2.0/24, send packets to 10.0.0.2 (R2's WAN IP)"
! 10.0.0.2 is the NEXT HOP — the IP of the directly connected neighbor

! % % % Default route (toward internet / upstream router) % % %
! If you had an ISP, the next-hop would be the ISP gateway IP
! For this lab, we skip the default route — both networks are known

end
copy running-config startup-config
```

2 Configure Router R2

R2 — Static Route Configuration

```
enable
configure terminal
hostname R2
no ip domain-lookup

interface GigabitEthernet0/0
description *** LAN — toward PC-B ***
ip address 10.2.2.1 255.255.255.0
no shutdown
exit

interface GigabitEthernet0/1
description *** WAN — toward R1 ***
ip address 10.0.0.2 255.255.255.252
no shutdown
exit

! % % % Static route back to R1's LAN network % % %
ip route 10.1.1.0 255.255.255.0 10.0.0.1
! Read as: "To reach 10.1.1.0/24, send packets to 10.0.0.1 (R1's WAN IP)"
! CRITICAL: BOTH routers need routes to reach each other's LANs
! If R1 has a route to R2's LAN but R2 doesn't have a route back,
! pings LEAVE PC-A and ARRIVE at PC-B — but the reply cannot get back.
! This is called ASYMMETRIC or ONE-WAY routing — it looks like packet loss.

end
copy running-config startup-config
```

3 Verify

Verification

```
! On R1 — view the full routing table
show ip route
! Expected output:
! Codes: C - connected, S - static, R - RIP, O - OSPF
!
! C 10.0.0.0/30 is directly connected, GigabitEthernet0/1
! L 10.0.0.1/32 is directly connected, GigabitEthernet0/1
! C 10.1.1.0/24 is directly connected, GigabitEthernet0/0
! L 10.1.1.1/32 is directly connected, GigabitEthernet0/0
! S 10.2.2.0/24 [1/0] via 10.0.0.2 !* Your static route!
! [1/0] means Administrative Distance 1 / Metric 0

! Ping test from PC-A ! PC-B
! (PC-A cmd prompt): ping 10.2.2.10
! Expected: Reply from 10.2.2.10 ...

! Traceroute to see path
! (PC-A cmd prompt): traceroute 10.2.2.10
! Expected: 1: 10.1.1.1 (R1) 2: 10.2.2.10 (PC-B)
```

COMMON ERROR

COMMON ERROR: Pings from PC-A timeout even though routes look correct.

CAUSE: PC-A does not have a default gateway set, OR R2 is missing the return static route.

FIX: (1) Check PC-A default gateway is 10.1.1.1. (2) Run "show ip route" on R2 and confirm 10.1.1.0/24 is

in its routing table.

Inter-VLAN Routing (Router-on-a-Stick)

50 minutes · + Intermediate

OBJECTIVES

- Understand why VLANs cannot communicate without Layer 3 routing
- Configure sub-interfaces on a router with 802.1Q encapsulation
- Enable inter-VLAN routing through a single physical interface
- Verify hosts in different VLANs can communicate through the router

CORE CONCEPTS — READ FIRST

THE INTER-VLAN PROBLEM: VLANs are isolated broadcast domains. A PC in VLAN 10 cannot reach a PC in VLAN 20 — they are in different networks (different IP subnets). Even if they share the same physical switch, the switch will not forward frames between VLANs. You NEED a Layer 3 device (router or Layer 3 switch) to route between VLANs.

ROUTER-ON-A-STICK: One physical router interface connected to a switch trunk port. The single interface is divided into logical sub-interfaces — one per VLAN. Each sub-interface has the 802.1Q tag for its VLAN and the IP address that serves as the default gateway for that VLAN. All VLAN traffic travels up the trunk to the router, gets routed between sub-interfaces, and comes back down.

SUB-INTERFACES: Logical divisions of a physical interface. The syntax is "interface Gi0/0.10" where Gi0/0 is the physical interface and 10 is the sub-interface number (by convention matches the VLAN ID, but not required). The "encapsulation dot1Q 10" command tells the sub-interface which tagged VLAN frames to accept. The physical parent interface must be up with "no shutdown" but has NO IP address itself.

ENCAPSULATION DOT1Q: Critical command on each sub-interface. "encapsulation dot1Q 10" means: "This sub-interface handles 802.1Q frames tagged with VLAN 10." Adding "native" at the end (encapsulation dot1Q 99 native) means this sub-interface handles untagged frames — use this for the native VLAN only.

TOPOLOGY

[illegible]

Ø=Ü; TIP

The physical interface Gi0/0 is like a highway on-ramp. The sub-interfaces (.10, .20, .99) are the individual lanes. The VLAN tag in each frame is like a license plate — it tells the router which "lane" (sub-interface) the frame belongs to.

1 Configure Switch SW1 — Trunk Uplink to Router

SW1 — Trunk toward Router

```
enable
configure terminal
hostname SW1

! Create VLANs
vlan 10
  name DATA
vlan 20
  name VOICE
vlan 99
  name NATIVE
exit

! Access ports
interface FastEthernet0/1
  switchport mode access
  switchport access vlan 10
  spanning-tree portfast
  no shutdown
exit

interface FastEthernet0/11
  switchport mode access
  switchport access vlan 20
  spanning-tree portfast
  no shutdown
exit

! Trunk port to ROUTER — this is key!
interface GigabitEthernet0/1
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,99
  switchport nonegotiate
  no shutdown
exit

end
copy running-config startup-config
```

2 Configure Router R1 — Sub-Interfaces

R1 — Router-on-a-Stick Sub-Interfaces

```
enable
configure terminal
hostname R1
no ip domain-lookup

! % % % CRITICAL: Physical interface must be UP with NO IP address % % %
interface GigabitEthernet0/0
  description *** TRUNK TO SW1 — Router-on-a-Stick ***
  no ip address
! Physical interface has NO IP — the sub-interfaces have IPs
no shutdown
! Parent interface MUST be "no shutdown" for sub-interfaces to work
exit

! % % % Sub-interface for VLAN 10 (Data) % % %
interface GigabitEthernet0/0.10
  description *** VLAN 10 DATA GATEWAY ***
  encapsulation dot1Q 10
! dot1Q 10 = this sub-interface accepts/sends 802.1Q frames tagged VLAN 10
  ip address 192.168.10.1 255.255.255.0
! This IP becomes the default gateway for all VLAN 10 hosts
exit

! % % % Sub-interface for VLAN 20 (Voice) % % %
interface GigabitEthernet0/0.20
  description *** VLAN 20 VOICE GATEWAY ***
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
exit

! % % % Sub-interface for VLAN 99 (Native — untagged traffic) % % %
interface GigabitEthernet0/0.99
  description *** NATIVE VLAN — Management ***
  encapsulation dot1Q 99 native
! The "native" keyword means: accept UNTAGGED frames on this sub-interface
  ip address 192.168.99.1 255.255.255.0
exit

end
copy running-config startup-config
```

3 Set PC IP Addresses

- PC-DATA: IP 192.168.10.10 / Mask 255.255.255.0 / Gateway 192.168.10.1
- PC-VOICE: IP 192.168.20.10 / Mask 255.255.255.0 / Gateway 192.168.20.1

4 Verify Inter-VLAN Routing

Verification

```
! On R1
show ip interface brief
! Expected: Gi0/0, Gi0/0.10, Gi0/0.20, Gi0/0.99 all show "up up"

show ip route
! Expected:
! C 192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
! C 192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
! C 192.168.99.0/24 is directly connected, GigabitEthernet0/0.99

! From PC-DATA — ping to PC-VOICE (different VLAN!)
ping 192.168.20.10
! Expected: Reply from 192.168.20.10 ...
! This works because: PC-DATA !' SW1 (VLAN 10 tag) !' Trunk !' R1 (routes)
! !' R1 Gi0/0.20 (VLAN 20 tag) !' Trunk !' SW1 !' PC-VOICE

! On SW1 — verify trunk is working
show interfaces GigabitEthernet0/1 trunk
! Verify native VLAN is 99 and VLANs 10,20,99 are active
```

COMMON ERROR

COMMON ERROR: PC-DATA can ping the router (192.168.10.1) but cannot ping PC-VOICE (192.168.20.10).

CAUSE: PC-VOICE does not have the correct default gateway set (must be 192.168.20.1, NOT 192.168.10.1).

FIX: Set PC-VOICE gateway to 192.168.20.1. Each host uses the gateway IP in its OWN subnet.

OBJECTIVES

- Understand why STP exists and what problem it solves
- Configure Rapid PVST+ and understand Rapid STP port states
- Manually elect the root bridge by setting bridge priority
- Configure PortFast and BPDUGuard on access ports
- Observe and test STP reconvergence

CORE CONCEPTS — READ FIRST

THE SWITCHING LOOP PROBLEM: Without STP, a network with redundant switch links creates a loop. A single broadcast frame (like an ARP request) gets forwarded around the loop indefinitely — never decreasing TTL at Layer 2. Within seconds, broadcast traffic fills 100% of all link bandwidth. Every frame is duplicated endlessly. This is called a broadcast storm and it crashes the entire network.

HOW STP WORKS: STP (and its faster successor RSTP) prevents loops by putting some switch ports into a BLOCKING state. Blocked ports receive STP messages but do NOT forward traffic. If the active path fails, STP detects the failure and unblocks a previously blocked port — but classic STP takes 30-50 seconds. Rapid STP (802.1w) converges in 1-2 seconds.

ROOT BRIDGE ELECTION: Every STP network has exactly ONE root bridge — the reference point for all path calculations. Root bridge is elected based on lowest Bridge ID = (Priority + VLAN ID) + MAC address. Default priority is 32768. ALWAYS manually set the root bridge — never leave it to chance. Configure the core or distribution switch with a low priority (4096 or 8192). If you leave defaults, the switch with the lowest MAC address wins — could be an old, weak switch.

PORT STATES (Rapid PVST+): Discarding — not forwarding, not learning (replaces Blocking+Listening). Learning — learning MACs but not forwarding. Forwarding — fully operational. Note: Classic 802.1D has 5 states and takes 30-50s. Rapid STP has 3 effective states and converges in ~1 second.

PORTFAST & BPDUGUARD: PortFast skips the Discarding! Learning! Forwarding sequence and immediately enters Forwarding state. Use ONLY on access ports connected to end devices — never on switch-to-switch links. BPDUGuard protects PortFast ports: if ANY BPDU is received on a PortFast port (meaning a switch was plugged in), the port immediately enters "err-disabled" state. This prevents STP manipulation attacks.

TOPOLOGY

[illegible]

1 Configure CORE-SW as Root Bridge

CORE-SW — Root Bridge Configuration

```
enable
configure terminal
hostname CORE-SW

! % % % Enable Rapid PVST+ on all switches % % %
spanning-tree mode rapid-pvst
! Rapid PVST+ = Per-VLAN rapid spanning tree
! One STP instance per VLAN — each VLAN can have a different topology

! % % % Set priority — CORE-SW is ROOT for ALL VLANs % % %
spanning-tree vlan 1,10,20,30,40,99 priority 4096
! Priority MUST be a multiple of 4096: 0, 4096, 8192, 12288, 16384...32768
! Default is 32768. Lower number = higher chance of being root.
! 4096 ensures CORE-SW wins the election against any default-priority switch.

! Alternatively use this shortcut command:
! spanning-tree vlan 10 root primary
! This automatically sets priority to 24576 or lower to beat current root.

! % % % Enable BPDUGuard globally on all PortFast ports % % %
spanning-tree portfast bpduguard default
! This applies BPDUGuard to ALL ports that have "spanning-tree portfast" configured.
! You do not need to configure it per-port separately.

! % % % Configure trunk uplinks to distribution layer % % %
interface GigabitEthernet0/1
description *** TRUNK TO DIST-A ***
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,30,40,99
spanning-tree guard root
! Root Guard: if a SUPERIOR BPDU arrives on this port (a switch claiming to
! be root), the port goes into "root-inconsistent" state — blocking it.
! Prevents unauthorized switches from becoming the root bridge.
no shutdown
exit

interface GigabitEthernet0/2
description *** TRUNK TO DIST-B ***
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,30,40,99
spanning-tree guard root
no shutdown
exit

end
copy running-config startup-config
```

2

Configure DIST-A and DIST-B

DIST-A / DIST-B Configuration

```
! % % % On DIST-A % % %
enable
configure terminal
hostname DIST-A

spanning-tree mode rapid-pvst
spanning-tree vlan 1,10,20,30,40,99 priority 8192
! DIST-A is secondary root — becomes root if CORE-SW fails

interface GigabitEthernet0/1
description *** TRUNK TO CORE-SW ***
switchport mode trunk
switchport trunk native vlan 99
spanning-tree guard root
no shutdown

interface GigabitEthernet0/2
description *** TRUNK TO ACCESS-SW-1 ***
switchport mode trunk
switchport trunk native vlan 99
no shutdown

end
copy running-config startup-config

! % % % On DIST-B (same but different priority) % % %
! hostname DIST-B
! spanning-tree vlan 1,10,20,30,40,99 priority 12288
```

3 Configure ACCESS-SW-1 — PortFast + BPDUGuard

ACCESS-SW-1 — PortFast Configuration

```
enable
configure terminal
hostname ACCESS-SW-1

spanning-tree mode rapid-pvst
! High priority — access switch should NEVER become root
spanning-tree vlan 1,10,20,30,40,99 priority 28672

! % % % Access ports — PortFast + BPDUGuard % % %
interface range FastEthernet0/1 - 20
  switchport mode access
  switchport access vlan 10
  spanning-tree portfast
! PortFast: port goes immediately to Forwarding state
! End device gets network access in <1 second instead of 30 seconds
  spanning-tree bpduguard enable
! If a switch is plugged into this port and sends a BPDU,
! the port goes err-disabled immediately
  no shutdown
  exit

! % % % Uplink ports to DIST-A and DIST-B % % %
interface GigabitEthernet0/1
  description *** TRUNK TO DIST-A (PRIMARY) ***
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  spanning-tree portfast disable
! NEVER enable PortFast on trunk/uplink ports!
  no shutdown
  exit

interface GigabitEthernet0/2
  description *** TRUNK TO DIST-B (REDUNDANT) ***
  switchport mode trunk
  switchport trunk native vlan 99
  spanning-tree portfast disable
  no shutdown
  exit

end
copy running-config startup-config
```

4 Verify STP

STP Verification

! On CORE-SW — should show "This bridge is the root"

```
show spanning-tree vlan 10
```

! Expected output (key lines):

! VLAN0010

! Spanning tree enabled protocol rstp

! Root ID Priority 4106 (4096 base + 10 VLAN)

! Address xxxx.xxxx.xxxx

! This bridge is the root !• MUST see this on core!

! Bridge ID Priority 4106

! On DIST-B — check which port is blocking (designated root port and alts)

```
show spanning-tree vlan 10
```

! Look for port role "Altn" and state "BLK" — this is the blocked port

! The blocked port breaks the loop but stays ready for failover

! View brief STP summary for all VLANs

```
show spanning-tree summary
```

! Verify PortFast and BPDUGuard on access switch

```
show spanning-tree interface FastEthernet0/1 portfast
```

! Expected: PVST enabled, port Fast: yes (auto), bpduguard: yes

! Test BPDUGuard (OPTIONAL — for demonstration):

! Plug a switch into Fa0/1 !' the port goes err-disabled within seconds

```
show interfaces FastEthernet0/1 status
```

! Status shows "err-disabled"

! To recover: remove the switch, then:

```
interface FastEthernet0/1
```

```
shutdown
```

```
no shutdown
```

EXPECTED RESULT

STP FAILOVER TEST: Disconnect the cable between CORE-SW and DIST-A. Watch the previously blocking port on ACCESS-SW-1 transition to Forwarding — with Rapid PVST+ this takes about 1 second. This is the redundancy in action. Reconnect the cable — CORE-SW reestablishes as root and the topology restores.

OSPF Single Area (Open Shortest Path First) + Intermediate

OBJECTIVES

- Understand OSPF neighbor relationships and the LSDB
- Configure OSPF Area 0 on multiple routers
- Set Router IDs and understand their importance
- Configure passive interfaces to protect internal networks
- Enable OSPF MD5 authentication to prevent unauthorized neighbors
- Redistribute a default route through OSPF

CORE CONCEPTS — READ FIRST

OSPF vs STATIC ROUTING: Static routes are manually configured and never change. OSPF is a dynamic routing protocol — routers discover each other, share topology information, and automatically calculate the best path using Dijkstra's shortest path algorithm. If a link fails, OSPF reroutes around it automatically within seconds. OSPF is the industry standard for enterprise interior routing.

HELLO PROTOCOL & NEIGHBOR RELATIONSHIPS: OSPF routers discover each other by sending Hello packets every 10 seconds (on Ethernet) to multicast address 224.0.0.5. Two routers become OSPF neighbors when their: Hello/Dead timers match, Area ID matches, Subnet matches, Authentication matches, MTU matches. The Dead interval is 4x Hello = 40 seconds. If no Hello is received for 40 seconds, the neighbor is declared dead and routes are removed.

LINK STATE DATABASE (LSDB): Unlike RIP which shares routing tables, OSPF routers share topology information (Link State Advertisements - LSAs). Each router builds a complete map of the network (the LSDB) and runs Dijkstra's algorithm on its own copy to calculate the best path to every network. All routers in an area have identical LSDBs.

OSPF AREAS: Large OSPF networks are divided into areas to limit the scope of LSA flooding and reduce routing table size. Area 0 (backbone area) is mandatory — all other areas must connect to Area 0. In this lab, we use a single Area 0 (simplest configuration). CCNA focuses primarily on single-area OSPF.

ROUTER ID: A 32-bit number that uniquely identifies each OSPF router — formatted like an IP address (e.g., 1.1.1.1). Selection order: (1) Manually configured with "router-id" command — ALWAYS do this. (2) Highest loopback IP. (3) Highest active interface IP. Manually set the Router ID to avoid unpredictable behavior when interfaces go up/down.

PASSIVE INTERFACES: By default, OSPF sends Hello packets out ALL interfaces in the network statement. This is wasteful and a security risk on LAN segments with no OSPF neighbors. "passive-interface" stops Hello packets on that interface but still advertises the connected network. Use "passive-interface default" then "no passive-interface" on links that actually have OSPF neighbors.

OSPF COST: The metric OSPF uses to choose the best path. $\text{Cost} = \text{Reference Bandwidth} / \text{Interface Bandwidth}$. Default reference bandwidth is 100 Mbps. FastEthernet (100M) = cost 1. GigabitEthernet (1000M) = also cost 1 by default! You must manually adjust: "ip ospf cost" per interface OR "auto-cost reference-bandwidth 1000" on all routers for gigabit accuracy.

[illegible]

Router	Interface	IP / Mask	Description
R1	Gi0/0	10.1.1.1/24	LAN-A (users)
R1	Gi0/1	10.0.12.1/30	WAN to R2
R1	Gi0/2	10.0.13.1/30	WAN to R3
R2	Gi0/0	10.2.2.1/24	LAN-B (users)
R2	Gi0/1	10.0.12.2/30	WAN to R1
R3	Gi0/0	10.3.3.1/24	LAN-C (branch)
R3	Gi0/1	10.0.13.2/30	WAN to R1

1 Configure R1 with OSPF

R1 — OSPF Configuration

```
enable
configure terminal
hostname R1
```

! % % % Configure interfaces first % % %

```
interface GigabitEthernet0/0
description *** LAN-A ***
ip address 10.1.1.1 255.255.255.0
no shutdown
exit
```

```
interface GigabitEthernet0/1
description *** WAN TO R2 ***
ip address 10.0.12.1 255.255.255.252
no shutdown
exit
```

```
interface GigabitEthernet0/2
description *** WAN TO R3 ***
ip address 10.0.13.1 255.255.255.252
no shutdown
exit
```

! % % % Configure OSPF % % %

```
router ospf 1
```

! "1" is the OSPF Process ID — local to this router only.

! R1 process 1 and R2 process 2 can still become OSPF neighbors.

! Process ID does NOT need to match between routers.

```
router-id 1.1.1.1
```

! Set Router ID FIRST, before network statements.

! If you change the RID later, you must use "clear ip ospf process" to apply it.

! % % % Advertise networks in OSPF Area 0 % % %

```
network 10.1.1.0 0.0.0.255 area 0
```

! Format: network [network-address] [wildcard-mask] area [area-id]

! Wildcard mask is the INVERSE of the subnet mask:

! /24 mask = 255.255.255.0, wildcard = 0.0.0.255

! /30 mask = 255.255.255.252, wildcard = 0.0.0.3

! /32 mask = 255.255.255.255, wildcard = 0.0.0.0

```
network 10.0.12.0 0.0.0.3 area 0
```

```
network 10.0.13.0 0.0.0.3 area 0
```

! % % % Passive interfaces (no OSPF hellos toward end users) % % %

```
passive-interface GigabitEthernet0/0
```

! Gi0/0 faces the LAN — no other OSPF routers there.

! passive-interface stops sending Hello packets but still ADVERTISES

! the 10.1.1.0/24 network to other OSPF neighbors.

! % % % Adjust reference bandwidth for Gigabit accuracy % % %

```
auto-cost reference-bandwidth 1000
```

! With default 100Mbps reference, both FastEthernet AND GigabitEthernet

! get cost=1 — OSPF cannot differentiate speed. With 1000Mbps:

! GigabitEthernet = cost 1, FastEthernet = cost 10. Must set on ALL routers.

exit



Configure R2 and R3 with OSPF

R2 and R3 — OSPF Configuration

```
! % % % R2 Configuration % % %
enable
configure terminal
hostname R2

interface GigabitEthernet0/0
 ip address 10.2.2.1 255.255.255.0
 no shutdown
interface GigabitEthernet0/1
 ip address 10.0.12.2 255.255.255.252
 no shutdown

router ospf 1
 router-id 2.2.2.2
 network 10.2.2.0 0.0.0.255 area 0
 network 10.0.12.0 0.0.0.3 area 0
 passive-interface GigabitEthernet0/0
 auto-cost reference-bandwidth 1000
 exit

end
copy running-config startup-config

! % % % R3 Configuration % % %
! (apply on R3 similarly)
! hostname R3
! router ospf 1
! router-id 3.3.3.3
! network 10.3.3.0 0.0.0.255 area 0
! network 10.0.13.0 0.0.0.3 area 0
! passive-interface GigabitEthernet0/0
! auto-cost reference-bandwidth 1000
```

3 Add OSPF MD5 Authentication (Security Hardening)

OSPF MD5 Authentication

! On ALL routers — Add MD5 authentication to WAN interfaces
! Prevents unauthorized routers from forming OSPF adjacencies

! On R1:

```
interface GigabitEthernet0/1
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 OspfKey@99
```

! Key ID (1) and password MUST match on both ends of the link

```
interface GigabitEthernet0/2
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 OspfKey@99
```

! On R2 Gi0/1 and R3 Gi0/1 — same commands
! WARNING: After adding auth, the neighbor relationship will DROP
! and re-form once BOTH sides have authentication configured.
! This is normal — wait 40 seconds for neighbors to reform.

4 Verify OSPF

OSPF Verification

! Check OSPF neighbor state — MUST be FULL

```
show ip ospf neighbor
```

! Expected:

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/DR	00:00:38	10.0.12.2	Gi0/1
3.3.3.3	1	FULL/DR	00:00:35	10.0.13.2	Gi0/2

! State FULL = adjacency complete, LSDBs synchronized

! Any other state (INIT, 2WAY, EXSTART, EXCHANGE, LOADING) = problem

! Check OSPF routes in routing table

```
show ip route ospf
```

! Expected on R1:

```
! O 10.2.2.0/24 [110/2] via 10.0.12.2, Gi0/1 !* R2 LAN
! O 10.3.3.0/24 [110/2] via 10.0.13.2, Gi0/2 !* R3 LAN
! [110/2] = AD 110 (OSPF default) / Metric (cost) 2
```

! View the OSPF database

```
show ip ospf database
```

! This shows all LSAs — all routers in area should have same database

! Ping end-to-end

! From PC-LAN-A: ping 10.2.2.10 (PC-LAN-B)

! Expected: Success

COMMON ERROR

COMMON ERROR: OSPF neighbors stuck in INIT or 2WAY state, never reaching FULL.

CAUSE #1: Subnet mismatch — e.g., one side is /30 and other is /29

CAUSE #2: Authentication mismatch — key ID or password differs

CAUSE #3: Hello/Dead timer mismatch

FIX: "show ip ospf interface Gi0/1" on BOTH routers — compare timers, auth type, and network type. All must match.

- Configure a Cisco router as a DHCP server
- Exclude static IP addresses from the DHCP pool
- Configure DHCP options: default gateway, DNS server, domain name, lease time
- Configure DHCP relay (ip helper-address) for inter-VLAN DHCP
- Verify DHCP leases and troubleshoot common issues

HOW DHCP WORKS (4-Way Handshake — DORA): Discover: Client broadcasts "Is there a DHCP server?" (destination 255.255.255.255). Offer: Server responds with an available IP address offer. Request: Client broadcasts "I accept that IP" (still broadcast — there may be multiple servers). Acknowledge: Server sends the final lease with IP, mask, gateway, DNS, and lease time. The client is now configured.

LEASE TIME: How long a client can use the assigned IP. Short leases (1 hour) waste bandwidth with frequent renewals but quickly reclaim unused IPs. Long leases (7 days) reduce DHCP traffic but can exhaust the pool if old devices leave without releasing. The client renews at 50% of the lease time (T1) and retries at 87.5% (T2). In a lab: use "lease 0 1" (1 hour) or "lease 1" (1 day).

TOPOLOGY

[illegible]

1 Configure DHCP Server on R1

R1 — DHCP Server Configuration

```
enable
configure terminal

! % % % STEP 1: Exclude static/reserved addresses FIRST % % %
! Always do exclusions BEFORE defining pools
ip dhcp excluded-address 192.168.10.1 192.168.10.20
! Excludes .1 through .20 — keeps them available for static assignment
! Routers, switches, printers, servers will use IPs in this range

ip dhcp excluded-address 192.168.20.1 192.168.20.20
ip dhcp excluded-address 192.168.30.1 192.168.30.20

! % % % STEP 2: Define DHCP pool for VLAN 10 (Data) % % %
ip dhcp pool VLAN10-DATA
  network 192.168.10.0 255.255.255.0
! The scope — IPs from this network are available (minus exclusions)
  default-router 192.168.10.1
! This is the default gateway sent to clients
! MUST match the actual gateway IP of VLAN 10
  dns-server 8.8.8.8 8.8.4.4
! Up to 8 DNS servers can be listed
  domain-name novatech.local
! Clients use this for local DNS resolution
  lease 1
! Lease time: 1 day
! Format: lease [days] [hours] [minutes]
! "lease 0 8" = 8 hours, "lease 7" = 7 days
  exit

! % % % Pool for VLAN 20 (Voice) % % %
ip dhcp pool VLAN20-VOICE
  network 192.168.20.0 255.255.255.0
  default-router 192.168.20.1
  dns-server 8.8.8.8
  option 150 ip 192.168.20.10
! Option 150 = TFTP server IP for Cisco IP phone firmware download
! VoIP phones use this to download their config from the call manager
  lease 1
  exit

! % % % Pool for VLAN 30 (Management) % % %
ip dhcp pool VLAN30-MGMT
  network 192.168.30.0 255.255.255.0
  default-router 192.168.30.1
  dns-server 8.8.8.8
  lease 0 12
! Shorter lease for management devices
  exit

end
copy running-config startup-config
```

2 Configure DHCP Relay on Distribution Switch (or Router SVI)

DHCP Relay — ip helper-address

! On the Layer 3 device that acts as the VLAN gateway
! (This could be a router sub-interface or L3 switch SVI)

! For Router sub-interface approach:

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.40.1
```

! ip helper-address = the IP of your DHCP server

! When a DHCP broadcast arrives on this interface:

- ! 1. Router intercepts the broadcast
- ! 2. Changes source IP to its own IP (192.168.10.1)
- ! 3. Forwards as unicast to DHCP server at 192.168.40.1
- ! 4. Server replies to 192.168.10.1 (the relay)
- ! 5. Relay forwards the offer back to the client

```
no shutdown
```

```
exit
```

```
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.40.1
 no shutdown
 exit
```

! For L3 Switch SVI approach:

! interface Vlan10

! ip address 192.168.10.1 255.255.255.0

! ip helper-address 192.168.40.1

! no shutdown

3 Verify DHCP

DHCP Verification

! On R1 — view all current DHCP bindings (leases)

```
show ip dhcp binding
```

! Expected:

! IP address	Client-ID/Hardware address	Lease expiration	Type
! 192.168.10.21	0100.50xx.xxxx.xx	Jun 04 2026	Automatic

! View DHCP pool statistics

```
show ip dhcp pool
```

! Shows: Total addresses, used, available in each pool

! Check for IP conflicts (addresses in use by non-DHCP devices)

```
show ip dhcp conflict
```

! Should be empty. If addresses appear here, they were offered

! but a conflict was detected (another device has that IP).

! Clear with: clear ip dhcp conflict *

! On a PC — verify it received an IP

! PC !' Desktop !' IP Configuration !' select DHCP

! PC should receive: IP in range .21-.254, GW .1, DNS 8.8.8.8

! Packet Tracer: Go to PC !' Desktop !' IP Configuration

! Switch from Static to DHCP and observe the IP being assigned

COMMON ERROR

COMMON ERROR: PC shows 169.254.x.x (APIPA address — Automatic Private IP Addressing).

MEANING: The PC did not receive a DHCP response. It assigned itself an APIPA address.

CAUSE: No route from PC to DHCP server, missing ip helper-address, or DHCP pool not defined.

FIX: (1) Check ip helper-address is configured on the gateway interface. (2) Check the DHCP pool "network" matches the VLAN subnet. (3) Verify the gateway interface is up ("show ip int brief").

OBJECTIVES

- Understand why NAT exists and the difference between Static NAT, Dynamic NAT, and PAT
- Configure Static NAT for a server that needs a fixed public IP
- Configure PAT (NAT overload) to give internet access to all internal hosts
- Define NAT inside and outside interfaces correctly
- Verify NAT translations and debug NAT issues

CORE CONCEPTS — READ FIRST

WHY NAT? The internet uses public IPv4 addresses, and there are only ~4.3 billion total. Private IP addresses (RFC1918: 10.x, 172.16-31.x, 192.168.x) cannot be routed on the public internet. NAT (Network Address Translation) translates between private and public addresses at the edge router, allowing thousands of internal hosts to share one (or a few) public IP addresses. Almost every home and enterprise network uses NAT.

STATIC NAT: One-to-one mapping. Private IP 192.168.40.10 always translates to public IP 203.0.113.50. Used for servers that must be reachable from the internet at a fixed address (web servers, mail servers). The translation exists permanently — even when no traffic is flowing.

PAT (Port Address Translation / NAT Overload): Many-to-one mapping. All internal private IPs translate to a SINGLE public IP, differentiated by port numbers. PC-A sends traffic from port 50123, PC-B from port 50456 — both map to the same public IP but different ports. The NAT table tracks these mappings and returns traffic to the correct internal host. This is what your home router does — all your devices share one public IP.

INSIDE vs OUTSIDE: Every interface must be declared as either "ip nat inside" (faces the private network) or "ip nat outside" (faces the public internet). NAT only translates packets that cross the inside-to-outside boundary. If you forget to set these, NAT translations will never be created, even if the NAT config looks correct.

NAT ACL: For PAT, you define an ACL that specifies WHICH private addresses to translate. Only traffic matching the ACL gets NAT'd. Traffic not matching the ACL passes through untranslated — which means it uses a private IP and gets dropped by the ISP. Always make sure your ACL covers all internal subnets that need internet access.

TOPOLOGY

LAB 08 TOPOLOGY			
INTERNAL (Private)	EDGE-RTR	INTERNET	
192.168.10.0/24 (VLAN10)	Gi0/0: Inside	Gi0/1: Outside	
192.168.20.0/24 (VLAN20)		% % % % ISP	
192.168.40.0/24 (Servers)		203.0.113.1	
	203.0.113.2/30		
Web Server: 192.168.40.10			
! ' Static NAT: 203.0.113.50			
All other hosts ! ' PAT			
(share 203.0.113.2)			

1 Configure NAT on EDGE-RTR

EDGE-RTR — NAT Configuration Part 1

```
enable
configure terminal
hostname EDGE-RTR

! % % % Step 1: Configure interfaces with inside/outside designation % % %
interface GigabitEthernet0/0
  description *** INSIDE — toward LAN ***
  ip address 192.168.1.1 255.255.255.0
  ip nat inside
! This interface faces the private network — NAT translates FROM here
no shutdown
exit

interface GigabitEthernet0/1
  description *** OUTSIDE — toward ISP/Internet ***
  ip address 203.0.113.2 255.255.255.252
  ip nat outside
! This interface faces the public internet — NAT translates TO here
no shutdown
exit

! % % % Step 2: Default route toward ISP % % %
ip route 0.0.0.0 0.0.0.0 203.0.113.1
! All unknown traffic goes to ISP

! % % % Step 3: Static NAT for Web Server % % %
ip nat inside source static 192.168.40.10 203.0.113.50
! Permanent one-to-one mapping:
! 192.168.40.10 (private) !•! 203.0.113.50 (public)
! Anyone on internet connecting to 203.0.113.50 !' gets forwarded to 192.168.40.10
! The server at 192.168.40.10 uses 203.0.113.50 as its "public face"
```


EDGE-RTR — NAT Configuration Part 2

! % % % Step 4: PAT (NAT Overload) for all internal hosts % % %

! First: Define WHICH internal networks get NAT'd

```
ip access-list standard NAT-INSIDE-NETWORKS
 permit 192.168.10.0 0.0.0.255
 permit 192.168.20.0 0.0.0.255
 permit 192.168.30.0 0.0.0.255
```

! Note: 192.168.40.0 (servers) uses Static NAT above — exclude from PAT

! Or include it — static NAT entries take priority over PAT for .40.10

! Apply PAT: all matched internal IPs !' translated to Gi0/1 public IP

```
ip nat inside source list NAT-INSIDE-NETWORKS interface GigabitEthernet0/1 overload
```

! "list NAT-INSIDE-NETWORKS" = use this ACL to identify inside traffic

! "interface GigabitEthernet0/1" = use the outside interface IP as the public IP

! "overload" = PAT — allow many-to-one with port tracking

! % % % Sub-interfaces if using Router-on-a-Stick % % %

! (If your internal network uses VLANs via sub-interfaces)

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 no shutdown
```

```
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip nat inside
 no shutdown
```

end

```
copy running-config startup-config
```

2 Verify NAT

NAT Verification

```
! View active NAT translations
show ip nat translations
! (Generate some traffic from internal PCs first)
! Expected output:
! Pro Inside global    Inside local    Outside local  Outside global
! --- 203.0.113.50      192.168.40.10   ---           ---
! icmp 203.0.113.2:1025 192.168.10.21:1025 8.8.8.8:1025 8.8.8.8:1025
! icmp 203.0.113.2:1026 192.168.10.22:1026 8.8.8.8:1026 8.8.8.8:1026
! Note: Two PCs share 203.0.113.2 but use different ports (1025, 1026)

! View NAT statistics
show ip nat statistics
! Look for:
! "Total active translations: X"
! "Outside interfaces: GigabitEthernet0/1"
! "Inside interfaces: Gi0/0.10, Gi0/0.20 ..."
! "Hits: X Misses: Y" — Misses means traffic not being translated

! From internal PC — ping internet (simulated server in Packet Tracer)
ping 8.8.8.8
! If success: NAT is working

! Debug NAT in real-time (careful — verbose output)
debug ip nat
! Then ping from a PC — you will see translation events
! Disable debug: "undebug all" or "no debug all"
```

COMMON ERROR

COMMON ERROR: Internal hosts can ping the router but NOT the internet.

CAUSE: Missing "ip nat inside" or "ip nat outside" on interfaces.

FIX: "show ip nat statistics" — check inside/outside interfaces are listed. If not, add "ip nat inside/outside" to the correct interfaces.

SECOND CAUSE: ACL in "ip nat inside source list" does not match the internal IP range. Check with "show access-lists".

- Understand the difference between Standard and Extended ACLs
- Write Standard ACLs to filter by source IP
- Write Extended ACLs to filter by source, destination, protocol, and port
- Apply ACLs in the correct direction (inbound vs outbound) and correct interface
- Create Named ACLs for readability and ease of editing

CORE CONCEPTS — READ FIRST

WHAT IS AN ACL? An Access Control List is an ordered list of permit/deny statements. A router checks each packet against the ACL entries in order, from top to bottom. When the first match is found, the action (permit or deny) is applied and the rest of the ACL is skipped. Every ACL ends with an IMPLICIT DENY ALL — any traffic not matched by an explicit permit is automatically denied. If your ACL only permits specific traffic, all other traffic is blocked.

STANDARD ACL (1-99 or named): Matches only on SOURCE IP address. Cannot specify destination, protocol, or port. Because of the limited matching, Standard ACLs are placed CLOSE TO THE DESTINATION — this prevents blocking traffic that should go to other destinations. Standard ACLs are good for: blocking a specific host from the whole network, permit only management VLAN to access router VTY lines.

EXTENDED ACL (100-199 or named): Matches on source IP, destination IP, protocol (TCP/UDP/ICMP), and source/destination port. Much more precise. Extended ACLs are placed CLOSE TO THE SOURCE — this prevents unwanted traffic from traversing the network unnecessarily.

WILDCARD MASKS: ACLs use wildcard masks (same as OSPF). A 0 bit = match this bit exactly. A 1 bit = ignore this bit. "0.0.0.0" with IP 192.168.10.5 = match EXACTLY 192.168.10.5 (host). "0.0.0.255" with IP 192.168.10.0 = match any host in 192.168.10.0/24. Shortcuts: "host 192.168.10.5" = "192.168.10.5 0.0.0.0". "any" = "0.0.0.0 255.255.255.255".

INBOUND vs OUTBOUND: When you apply an ACL to an interface, you specify "in" or "out". "in" = packets ARRIVING on that interface are checked BEFORE routing. "out" = packets LEAVING that interface are checked AFTER routing. One ACL per direction per interface. Getting the direction wrong is the most common ACL configuration mistake.

TOPOLOGY

LAB 09 TOPOLOGY																																																	%	
																																																	%	
HR_PC: 192.168.10.5										R1										Server: 192.168.40.10																														
VLAN 10																				Gi0/0.10										VLAN 40																				%
										Gi0/0.20										VLAN 20 (Voice)																														%
										Gi0/0.40										VLAN 40 (Servers)																														%
GUEST_PC: 192.168.30.x										Gi0/1										ISP (NAT outside)																														%
VLAN 30																				Gi0/0.30																														%
																																																		%
GOALS:																																																	%	
1. Block GUEST (VLAN 30) from accessing Servers (VLAN 40)																																																		%
2. Only allow SSH to router from MGMT (VLAN 30)																																																		%
3. Block internet access from Server VLAN																																																		%
4. Only allow HR_PC to HR Database server (port 1433)																																																		%

1 Standard ACL — Restrict VTY (SSH) Access to Management VLAN Only

Standard ACL — VTY Protection

```
! GOAL: Only devices in 192.168.30.0/24 (Management VLAN) can SSH to the router
! Standard ACL — source IP only — place CLOSE TO DESTINATION (the router VTY)

enable
configure terminal

! Named Standard ACL (easier to read and edit than numbered)
ip access-list standard ACL-SSH-MGMT-ONLY
  permit 192.168.30.0 0.0.0.255
! Permit the entire management subnet (192.168.30.1 to .254)
  deny any log
! Deny everything else + log the denied attempts
! The "log" keyword generates a syslog message for every denied packet
exit
```

```
! Apply to VTY lines — inbound (packets arriving on VTY lines)
line vty 0 15
  access-class ACL-SSH-MGMT-ONLY in
! "access-class" is used for VTY lines (not "ip access-group")
! "access-class" applies to SOURCE IP of the SSH/Telnet session
exit
```

```
! TEST: SSH from management PC (192.168.30.x) ! SUCCESS
! TEST: SSH from data PC (192.168.10.x) ! REFUSED
```

2 Extended Named ACL — Block Guest from Server VLAN

Extended ACL — Block Guest from Servers

! GOAL: Block VLAN 30 (Guest 192.168.30.0/24) from reaching Servers (192.168.40.0/24)

! ALLOW all other traffic (VLAN 10, VLAN 20 can still reach servers)

! Extended ACL — matches source AND destination — place CLOSE TO SOURCE

```
ip access-list extended ACL-BLOCK-GUEST-SERVERS
deny ip 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255 log
```

! Deny ALL IP protocols (TCP, UDP, ICMP) from guest to server subnet

```
permit ip any any
```

! Permit everything else — without this, implicit deny blocks everything!

```
exit
```

! Apply to Gi0/0.30 INBOUND (as guests leave their VLAN toward router)

```
interface GigabitEthernet0/0.30
```

```
ip access-group ACL-BLOCK-GUEST-SERVERS in
```

! "in" = check packets ARRIVING from guests before routing

```
exit
```

! WHY inbound on .30? Extended ACL should be close to SOURCE.

! We block guest traffic RIGHT AS IT ENTERS the router from guest VLAN.

! This means guest traffic never wastes router processing reaching .40.

3 Extended ACL — Block Server VLAN from Internet

Extended ACL — Internet Access Policy

! GOAL: Servers (192.168.40.0/24) should NOT initiate outbound internet connections

! Servers should only RESPOND to incoming connections (return traffic allowed)

```
ip access-list extended ACL-INTERNET-POLICY
! Allow return/established TCP traffic from internet back to all internal hosts
permit tcp any 192.168.0.0 0.0.255.255 established
```

! "established" = matches TCP packets with ACK or RST flag set

! This allows RETURN traffic for sessions initiated by internal hosts

```
! Allow ICMP from ISP to edge router only (for monitoring)
```

```
permit icmp host 203.0.113.1 host 203.0.113.2
```

```
! Block Server VLAN (192.168.40.0/24) from initiating internet connections
```

```
deny ip 192.168.40.0 0.0.0.255 any log
```

```
! Allow all other internal traffic toward internet
```

```
permit ip 192.168.0.0 0.0.255.255 any
```

```
! Implicit deny all — blocks everything else
```

```
exit
```

! Apply OUTBOUND on the internet-facing interface

```
interface GigabitEthernet0/1
```

```
ip access-group ACL-INTERNET-POLICY out
```

! "out" = check packets LEAVING toward internet after routing decision

```
exit
```

4 Extended ACL — Permit Only HR PC to SQL Server

Extended ACL — HR Database Access Control

! GOAL: Only HR_PC (192.168.10.5) can reach SQL server at 192.168.40.10 on port 1433

! Other hosts in VLAN 10 cannot access the SQL server

```
ip access-list extended ACL-HR-DB-ACCESS
  permit tcp host 192.168.10.5 host 192.168.40.10 eq 1433
! "host 192.168.10.5" = exact single host (same as 192.168.10.5 0.0.0.0)
! "eq 1433" = destination port 1433 (Microsoft SQL Server)
! Common port numbers: 22=SSH, 23=Telnet, 25=SMTP, 80=HTTP, 443=HTTPS
!           53=DNS, 67/68=DHCP, 110=POP3, 443=HTTPS
!           1433=MSSQL, 3306=MySQL, 5432=PostgreSQL

! Allow established TCP (return traffic from SQL server back to HR PC)
permit tcp host 192.168.40.10 host 192.168.10.5 established

! Permit all other traffic from VLAN 10 (other users still work normally)
permit ip 192.168.10.0 0.0.0.255 any

deny ip any any log
exit

! Apply on the interface toward the server VLAN — outbound
interface GigabitEthernet0/0.40
  ip access-group ACL-HR-DB-ACCESS out
exit

end
copy running-config startup-config
```

5 Verify ACLs

ACL Verification

! View all ACLs and their match counters

```
show access-lists
```

! Expected:

! Standard IP access list ACL-SSH-MGMT-ONLY

! 10 permit 192.168.30.0, wildcard bits 0.0.0.255 (X matches)

! 20 deny any log (Y matches)

! Extended IP access list ACL-BLOCK-GUEST-SERVERS

! 10 deny ip 192.168.30.0 ... (Z matches)

! 20 permit ip any any

! View ACL applied to a specific interface

```
show ip interface GigabitEthernet0/0.30
```

! Look for: "Inbound access list is ACL-BLOCK-GUEST-SERVERS"

! Connectivity tests:

! From GUEST PC (192.168.30.x): ping 192.168.40.10 !' SHOULD FAIL

! From DATA PC (192.168.10.x): ping 192.168.40.10 !' SHOULD SUCCEED

! From HR_PC (192.168.10.5): connect to 192.168.40.10:1433 !' SHOULD SUCCEED

! From OTHER DATA PC: connect to 192.168.40.10:1433 !' SHOULD FAIL

& WARNING

ACL GOLDEN RULES:

1. Standard ACL !' Place CLOSE TO DESTINATION
2. Extended ACL !' Place CLOSE TO SOURCE

3. Every ACL has an implicit "deny any" at the end — always add explicit "permit ip any any" if needed
4. ACL entries are matched TOP TO BOTTOM — order matters. Put most specific entries first.
5. You cannot re-order entries in a numbered ACL. Delete and recreate, or use named ACLs with sequence numbers.

OBJECTIVES

- Understand the problem HSRP solves and how it works
- Configure HSRP v2 on two distribution switches
- Set priority and preempt to control which device is Active
- Configure HSRP interface tracking to trigger failover on uplink failure
- Test and observe HSRP failover in real time

CORE CONCEPTS — READ FIRST

THE SINGLE GATEWAY PROBLEM: End devices have one default gateway — one IP address. If that device (a router or L3 switch) goes down, ALL devices in that VLAN lose their internet connection — even if the network has redundant paths. The devices keep trying to send traffic to a MAC address that no longer responds. There is no automatic failover unless you use FHRP (First Hop Redundancy Protocol).

HOW HSRP WORKS: Two or more routers share a VIRTUAL IP and VIRTUAL MAC address. End devices use the virtual IP as their gateway. The ACTIVE router owns the virtual IP and responds to it. The STANDBY router monitors the active via hello packets (every 3 seconds). If no hello arrives for the holdtime (10 seconds), standby becomes active and takes over the virtual IP and MAC. End devices detect nothing — they continue sending to the same virtual MAC. Failover is seamless.

HSRP PRIORITY & PREEMPT: Default priority is 100. The router with HIGHER priority becomes active. If priorities are equal, the higher IP address wins. "standby preempt" means: if a higher-priority router comes back online after a failure, it immediately takes back the Active role. Without preempt, the current active keeps the role even if a higher-priority router returns. Always enable preempt on the intended primary router.

HSRP INTERFACE TRACKING: HSRP can monitor an interface state. If the tracked interface (typically the uplink to core) goes down, HSRP decrements the priority by a specified amount. If priority drops below the standby's priority, the standby takes over — even though the switch itself is still running. This ensures the device with the working uplink serves as the active gateway. Without tracking, the active could be a switch with a broken uplink — all traffic goes to it but cannot leave.

HSRP v1 vs v2: HSRPv1 uses multicast 224.0.0.2, group range 0-255, virtual MAC 0000.0c07.acXX. HSRPv2 uses multicast 224.0.0.102, group range 0-4095, virtual MAC 0000.0c9f.fXXX. Always use version 2 — it is more flexible and supports IPv6. Specify "standby version 2" on all devices in the group.

TOPOLOGY

[illegible]

1 Configure DIST-A as HSRP Active

DIST-A — HSRP Active Configuration

```
enable
configure terminal
hostname DIST-A

! Enable IP routing (required on L3 switches)
ip routing

! % % % VLAN configuration % % %
vlan 10
  name DATA
vlan 20
  name VOICE
vlan 99
  name NATIVE
exit

! % % % SVI for VLAN 10 with HSRP v2 % % %
interface Vlan10
  description *** DATA VLAN — HSRP ACTIVE ***
  ip address 192.168.10.2 255.255.255.0
! DIST-A physical IP is .2 — end devices use virtual IP .1

  standby version 2
! MUST specify version 2 — must match on all HSRP group members

  standby 10 ip 192.168.10.1
! Group 10 (matches VLAN 10 for clarity), Virtual IP = 192.168.10.1
! ALL devices in VLAN 10 use .1 as default gateway

  standby 10 priority 110
! Higher priority = more likely to become Active (default is 100)
! DIST-A wins with 110 vs DIST-B at 90

  standby 10 preempt delay minimum 15
! Preempt: if this router goes down and comes back, it re-claims Active role
! Delay: wait 15 seconds after coming back up before preempting
! (Gives OSPF/routing time to converge before HSRP takes over)

  standby 10 authentication md5 key-string HsrpKey@10
! MD5 authentication — prevents unauthorized HSRP devices in the group
! Key MUST match on DIST-A and DIST-B

  standby 10 track 1 decrement 30
! If tracked object 1 goes down: reduce priority by 30
! 110 - 30 = 80, which is less than DIST-B's 90 ! DIST-B takes over

no shutdown
exit

! % % % Interface tracking — monitor uplink to core % % %
track 1 interface GigabitEthernet0/1 line-protocol
! Track whether Gi0/1 (uplink to CORE-SW) is up
! When Gi0/1 goes down ! tracked object state changes ! HSRP decrements priority

! % % % HSRP for VLAN 20 (same pattern) % % %
```


! % % % Trunk interfaces % % %

Configure DIST-B as HSRP Standby

DIST-B — HSRP Standby Configuration

```
enable
configure terminal
hostname DIST-B
ip routing

vlan 10
  name DATA
vlan 20
  name VOICE
vlan 99
  name NATIVE
exit

interface Vlan10
  description *** DATA VLAN — HSRP STANDBY ***
  ip address 192.168.10.3 255.255.255.0
  ! DIST-B physical IP is .3

  standby version 2
  standby 10 ip 192.168.10.1
  ! Same virtual IP as DIST-A — they SHARE this VIP

  standby 10 priority 90
  ! Lower priority ! starts as Standby

  standby 10 preempt
  ! Preempt without delay — if DIST-B somehow beats DIST-A priority
  ! (would only happen if DIST-A tracks interface and decrements below 90)

  standby 10 authentication md5 key-string HsrpKey@10
  ! Authentication MUST match DIST-A exactly

  no shutdown
exit

interface Vlan20
  ip address 192.168.20.3 255.255.255.0
  standby version 2
  standby 20 ip 192.168.20.1
  standby 20 priority 90
  standby 20 preempt
  standby 20 authentication md5 key-string HsrpKey@20
  no shutdown
exit

interface GigabitEthernet0/1
  description *** UPLINK TO CORE-SW ***
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 99
  no shutdown

interface GigabitEthernet0/2
  description *** TRUNK TO ACCESS-SW ***
  switchport trunk encapsulation dot1q
```


Verify HSRP

HSRP Verification and Failover Test

! On DIST-A — verify it is Active

```
show standby brief
```

! Expected:

! P indicates configured to preempt.

! Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
! V10	10	110	P	Active	local	192.168.10.3	192.168.10.1
! V120	20	110	P	Active	local	192.168.20.3	192.168.20.1

! On DIST-B — verify it is Standby

```
show standby brief
```

! Expected:

! V10	10	90	P	Standby	192.168.10.2	local	192.168.10.1
-------	----	----	---	---------	--------------	-------	--------------

! Detailed HSRP info

```
show standby vlan 10
```

! Shows: hello/hold timers, virtual MAC, preempt state, authentication

! % % % FAILOVER TEST % % %

! Step 1: Start a continuous ping from PC1 to internet/gateway

! PC1 cmd: ping 192.168.10.1 -t (continuous in real OS)

! PC1 Packet Tracer: ping 192.168.10.1

! Step 2: Shut down DIST-A uplink to CORE-SW

```
interface GigabitEthernet0/1
shutdown
```

! Step 3: Watch HSRP state on DIST-A — should change from Active to Speak/Standby

```
show standby brief
```

! Priority drops: 110 - 30 = 80 (track decrement takes effect)

! Step 4: On DIST-B — it detects DIST-A missed hellos !' becomes Active

```
show standby brief
```

! DIST-B shows Active, DIST-A shows Standby (or Init if fully disconnected)

! Step 5: Re-enable DIST-A uplink

```
interface GigabitEthernet0/1
no shutdown
```

! After 15 seconds delay: DIST-A preempts !' becomes Active again

🔑 TIP

HSRP FAILOVER TIMING:

- Hello interval: 3 seconds (HSRPv2 default)
 - Holdtime: 10 seconds (3x hello)
 - Total failover time: up to 10 seconds in the worst case
 - With "standby preempt delay minimum 15": DIST-A waits 15 seconds after recovery before preempting
- This gives time for OSPF routes and STP to fully converge before traffic switches back.

OBJECTIVES

- Combine ALL previous lab technologies into a single enterprise topology
- Build a 3-tier hierarchical LAN (Core !' Distribution !' Access)
- Configure OSPF routing between HQ and a remote branch office
- Implement HSRP at the distribution layer for gateway redundancy
- Apply NAT/PAT for internet access at the edge
- Secure the access layer with DHCP Snooping, DAI, and Port Security
- Restrict management access and internet policy with ACLs
- Verify full end-to-end connectivity across all VLANs and sites
- Screenshot final topology as evidence for LinkedIn portfolio

ENTERPRISE SCENARIO

Company: NovaTech Solutions — A mid-sized technology firm

Requirement: Connect HQ (main office) to a branch office over a WAN link. Provide internet access to all users. Segregate Data, Voice, Server, and Management traffic into separate VLANs. Ensure that no single device failure disrupts network access for users.

COMPLETE TOPOLOGY DIAGRAM

```
% % % % % % % % % % % % %
%   INTERNET   %   ISP: 203.0.113.1/30
% % % % % % %, % % % % % %
%   Public: 203.0.113.2/30
% % % % % % %4% % % % % %
%   HQ-EDGE-RTR%   R1 — Edge Router
%   1.1.1.1   %   NAT, OSPF, DHCP, ACL
% % % % % % %, % % % % % %
%   Gi0/1 WAN   %   Gi0/2 Trunk to Core
10.0.0.1/30   %   172.16.0.1/30
%
% % % % % % %4% % % % % %
%   BRANCH-RTR %
%   6.6.6.6   %
%   10.0.0.2/30 %
% % % % % % %, % % % % % %
%   Trunk
% % % % % % %4% % % % % %
%   BRANCH-SW %
%   VLAN50   %
% % % % % % %, % % % % % %
Users: 192.168.50.x

% % % % % % % % % % % % %
% % % % % % %4% % % % % %
%   HQ-CORE-SW % (L3)
%   2.2.2.2   %
% % % % % % %, % % % % % %
%   Gi1/0/1 %   Gi1/0/2
%
% % % % % % %4% % % % % %4% % % % % %
%   DIST-A %   DIST-B %
%   3.3.3.3 %   4.4.4.4 %
%   HSRP Act %   HSRP Stby %
% % % % % %, % % % % % % % % % %
% % % % % % %, % % % % % %
% % % % % % %4% % % % % %
%   ACCESS-SW-1 %
%   VLAN 10,20 %
% % % % % % % % % % % % %
```

IP ADDRESS PLAN

HQ-EDGE-RTR

Gi0/0

ISP Uplink (NAT Outside)



HQ-EDGE-RTR

Gi0/1

HQ-EDGE-RTR

Gi0/2

[Routed link to Core](#)

HQ-CORE-SW

Vlan10 SVI

192.168.10.1/24

HQ-CORE-SW

Vlan20 SVI

HQ-CORE-SW

Vlan30 SVI

192.168.30.1/24

Mgmt VLAN Gateway

HQ-CORE-SW

Vlan40 SVI

192.168.40.1/24

DIST-A

Vlan10 SVI

DIST-B

Vlan10 SVI

192.168.10.3/24

BRANCH-RTR

Gi0/0

WAN to HQ

BRANCH-RTR

Gi0/1.50

192.168.50.1/24

VLAN	Name	Subnet	DHCP Range	Purpose
10	DATA	192.168.10.0/24	.21 – .254	Office PCs
20	VOICE	192.168.20.0/24	.21 – .254	VoIP Phones
30	MANAGEMENT	192.168.30.0/24	.21 – .254	Network devices
40	SERVERS	192.168.40.0/24	Static only	Application servers
50	BRANCH	192.168.50.0/24	.21 – .254	Branch office users
99	NATIVE	N/A	N/A	Trunk native VLAN

TECHNOLOGIES CHECKLIST

- ' VLANs 10, 20, 30, 40, 50, 99 — named and created on all switches
- ' 802.1Q Trunking — native VLAN 99 on all trunk links
- ' Rapid PVST+ — CORE-SW as root bridge (priority 4096)
- ' PortFast + BPDUGuard — on all access ports at ACCESS-SW-1
- ' OSPF Area 0 — all routers and L3 switches, MD5 auth on WAN links
- ' HSRP v2 — DIST-A active (priority 110), DIST-B standby (priority 90)
- ' HSRP Tracking — decrement 30 on uplink failure
- ' DHCP — pools for all VLANs, relay on gateway SVIs
- ' NAT/PAT — all internal subnets NATed at HQ-EDGE-RTR
- ' SSH v2 — on all devices, Telnet disabled
- ' ACL — Management-only VTY, internet policy, guest isolation
- ' DHCP Snooping — on all switches, VLANs 10,20,40
- ' Dynamic ARP Inspection — VLANs 10,20,40
- ' Port Security — max 2 MACs per access port, restrict violation
- ' Storm Control — 20% broadcast threshold

1 Build in Packet Tracer — Device List

- 1x Cisco 1941 or 2911 Router — HQ-EDGE-RTR
- 1x Cisco 1941 or 2911 Router — BRANCH-RTR
- 1x Cisco 3650 or 3560 L3 Switch — HQ-CORE-SW
- 2x Cisco 3650 or 3560 L3 Switch — DIST-A, DIST-B
- 1x Cisco 2960 Switch — ACCESS-SW-1
- 1x Cisco 2960 Switch — BRANCH-SW
- 4x PCs — PC-DATA, PC-VOICE, PC-SERVER, BRANCH-PC
- 1x Server (optional, for web/DNS testing)

2 Cabling Table — Connect All Devices

From Device	From Port	To Device	To Port	Cable Type
HQ-EDGE-RTR	Gi0/0	ISP (cloud/router)	Gi0/0	Straight-Through
HQ-EDGE-RTR	Gi0/1	BRANCH-RTR	Gi0/0	Crossover or Auto
HQ-EDGE-RTR	Gi0/2	HQ-CORE-SW	Gi1/0/24	Straight-Through
HQ-CORE-SW	Gi1/0/1	DIST-A	Gi0/1	Crossover or Auto
HQ-CORE-SW	Gi1/0/2	DIST-B	Gi0/1	Crossover or Auto

DIST-A

Gi0/2

ACCESS-SW-1

DIST-B

ACCESS-SW-1

DIST-A

DIST-B

ACCESS-SW-1

Fa0/1

PC-DATA

Straight-Through

ACCESS-SW-1

PC-VOICE

Straight-Through

BRANCH-RTR

Gi0/1

BRANCH-SW

Straight-Through

BRANCH-SW

Fa0/1

BRANCH-PC

Straight-Through

In Packet Tracer, use "Automatically Choose Connection Type" (the lightning bolt cable icon) if you are unsure which cable type to use. Packet Tracer will select the correct cable. In real life: routers use straight-through to switches, switches use crossover to switches — though modern Gigabit interfaces with Auto-MDIX handle both.

3 Apply All Configurations

Apply configurations in this **ORDER** to avoid dependency issues:

- 1. HQ-CORE-SW — VLANs, STP (root bridge), SVIs, OSPF, DHCP Snooping
- 2. DIST-A — VLANs, STP, HSRP Active, OSPF, Trunks
- 3. DIST-B — VLANs, STP, HSRP Standby, OSPF, Trunks
- 4. ACCESS-SW-1 — VLANs, Access Ports, PortFast, BPDUGuard, Port Security
- 5. HQ-EDGE-RTR — Interfaces, OSPF, NAT/PAT, DHCP Pools, ACLs, SSH
- 6. BRANCH-RTR — WAN Interface, OSPF, DHCP Pools, SSH
- 7. BRANCH-SW — VLANs, Trunk to BRANCH-RTR, Access Ports
- 8. PCs — Set to DHCP mode and verify they receive IP addresses

Use the configuration files from the `cisco-enterprise-project/configs/` folder you downloaded earlier. Each file contains the exact commands for its device.

4 Final Verification Checklist

Capstone Final Verification

! % % CHECKLIST: Run each command and confirm expected result % %

! 1. VLAN verification (on any switch)

`show vlan brief`

! Confirm: VLANs 10,20,30,40,99 exist with correct names and active ports

! 2. STP root bridge verification (on CORE-SW)

`show spanning-tree vlan 10`

! Confirm: "This bridge is the root" on CORE-SW

! 3. OSPF neighbors (on HQ-EDGE-RTR)

`show ip ospf neighbor`

! Confirm: All neighbors in FULL state (2.2.2.2, 3.3.3.3, 4.4.4.4, 6.6.6.6)

! 4. HSRP state (on DIST-A)

`show standby brief`

! Confirm: DIST-A shows Active for VLANs 10,20,30,40

! 5. DHCP bindings

`show ip dhcp binding`

! Confirm: PCs have received IPs in correct VLAN ranges

! 6. NAT translations

`show ip nat translations`

! Confirm: Active translations exist when PCs ping internet

! 7. End-to-end ping test

! PC-DATA (192.168.10.x) !' ping 192.168.50.x (BRANCH-PC) !• cross-site

! PC-DATA !' ping 192.168.20.1 (Voice GW) !• inter-VLAN

! PC-DATA !' ping 8.8.8.8 !• internet via NAT

! BRANCH-PC !' ping 192.168.40.10 (server) !• branch to HQ server

! 8. SSH test (from MGMT VLAN PC)

! SSH to 192.168.30.2 (CORE-SW) — should succeed

! SSH from DATA VLAN — should fail (ACL blocks it)

! 9. HSRP failover test

! Disconnect DIST-A uplink to CORE-SW

! Verify DIST-B becomes Active (show standby brief on DIST-B)

! Reconnect !' DIST-A preempts back after 15s

LINKEDIN PORTFOLIO SCREENSHOTS TO TAKE

- Ø=ÜØ Full Packet Tracer topology with all links green
- Ø=ÜØ "show ip ospf neighbor" on HQ-EDGE-RTR showing all FULL adjacencies
- Ø=ÜØ "show standby brief" on DIST-A showing Active state
- Ø=ÜØ "show vlan brief" on ACCESS-SW-1 showing all VLANs and ports
- Ø=ÜØ "show ip nat translations" showing multiple internal IPs sharing one public IP
- Ø=ÜØ Successful ping from BRANCH-PC to HQ Server (cross-site connectivity)
- Ø=ÜØ Failed ping proving ACL is working (guest VLAN blocked from servers)
- Ø=ÜØ HSRP failover: DIST-B becoming Active after DIST-A uplink shutdown

EXPECTED RESULT

LINKEDIN POST TIMING: Post your screenshots within 24 hours while the lab is fresh. Use the full LinkedIn article from the docs/linkedin-article.md file as your article. Post the shorter version as a regular post linking to the full article. Screenshots of working "show" commands are more credible than screenshots of topology diagrams alone.

APPENDIX A — QUICK REFERENCE COMMAND CHEAT SHEET

MODES — GETTING WHERE YOU NEED TO BE

Navigation

```
Router>           !• User exec mode (very limited)
Router> enable     !• Enter privileged exec
Router#           !• Privileged exec (show commands, ping, debug)
Router# conf t     !• Enter global config
Router(config)#    !• Global config (hostname, routing, VLANs)
Router(config)# interface Gi0/0 !• Enter interface config
Router(config-if)# !• Interface config (ip address, no shutdown)
Router(config-if)# exit         !• Go back to global config
Router(config)# end             !• Go back to privileged exec
Router# copy run start          !• SAVE the configuration!
```

SHOW COMMANDS — YOUR MOST USED TOOLS

Command	What it shows
show running-config	Current config in RAM (not saved yet)
show startup-config	Saved config in NVRAM
show ip interface brief	All interfaces: IP, status (up/down)
show interfaces	Detailed interface stats and errors
show vlan brief	VLAN database and port assignments
show interfaces trunk	Active trunk links and allowed VLANs
show ip route	Full routing table
show ip ospf neighbor	OSPF neighbor state (must be FULL)
show ip ospf database	OSPF LSDB — topology information
show standby brief	HSRP state for all groups
show ip dhcp binding	Active DHCP leases
show ip dhcp conflict	IP conflicts detected by DHCP
show ip nat translations	Active NAT/PAT translation table
show ip nat statistics	NAT hits/misses and interface roles
show access-lists	ACL entries and match counters
show spanning-tree vlan 10	STP topology for VLAN 10
show spanning-tree summary	STP summary across all VLANs
show port-security	Port security status all ports
show ip ssh	SSH version and state
show cdp neighbors detail	Directly connected Cisco devices
show version	IOS version, uptime, hardware info

APPENDIX B — COMMON ERRORS MASTER LIST

Every error you will likely encounter in these labs, the root cause, and the exact fix:

' ERROR: Invalid input detected at ^ marker

CAUSE: Wrong config mode, or IOS does not support the command on this device type

FIX: Check your prompt. "show ?" to see available commands in current mode. Some commands work only on L3 switches, not 2960s.

' ERROR: OSPF neighbors stuck in INIT

CAUSE: Router can hear hellos from neighbor but neighbor cannot hear its hellos — often an interface/ACL issue

FIX: "show ip ospf interface Gi0/1" — check area, auth, timers match on both sides.

' ERROR: OSPF neighbors stuck in 2WAY

CAUSE: On a broadcast segment with no DR elected — normal if both are DROTHER. On point-to-point, this is an error.

FIX: For p2p WAN links: "ip ospf network point-to-point" on both sides. 2WAY becomes FULL immediately.

' ERROR: OSPF neighbors drop after adding auth

CAUSE: Auth config on only one side — the configured side sees auth mismatch from the unconfigured side

FIX: Add auth to both sides quickly. Neighbors reform within 40 seconds once both sides match.

' ERROR: Native VLAN mismatch CDP warning

CAUSE: Trunk ends have different native VLANs configured

FIX: "switchport trunk native vlan 99" on BOTH ends of the trunk.

' ERROR: Ping fails — timeout, no reply

CAUSE: Missing route, missing return route, or wrong default gateway on PC

FIX: Traceroute to find where it stops. "show ip route" on the router at failure point. Check PC gateway.

' ERROR: PC shows 169.254.x.x address

CAUSE: DHCP request not reaching server (no relay, no route, or wrong pool)

FIX: Check ip helper-address on gateway SVI. Verify DHCP pool network matches VLAN subnet.

' ERROR: Cannot SSH — connection refused

CAUSE: "transport input telnet" instead of ssh, or SSH not enabled

FIX: "ip ssh version 2" and "transport input ssh" on VTY lines. Ensure RSA key generated.

' ERROR: Crypto key generate fails

CAUSE: hostname or ip domain-name not set before generating key

FIX: Set hostname and ip domain-name first, THEN generate RSA key.

' ERROR: BPDUGuard err-disabled port

CAUSE: A switch was plugged into a PortFast access port

FIX: Remove the switch. Then: interface FaX/X ! shutdown ! no shutdown.

' ERROR: ACL blocks too much traffic

CAUSE: Missing "permit ip any any" at end of extended ACL — implicit deny catches everything

FIX: Add "permit ip any any" as the last line. Check ACL with "show access-lists" for match counters.

' ERROR: NAT translations not created

CAUSE: "ip nat inside/outside" missing on interfaces

FIX: "show ip nat statistics" — verify inside/outside interfaces are listed. Add missing commands.

' ERROR: HSRP stays in LISTEN or SPEAK

CAUSE: Authentication mismatch between DIST-A and DIST-B

FIX: "show standby vlan 10" — check authentication type. Key must match exactly on both devices.

' ERROR: HSRP virtual IP not reachable

CAUSE: PC has wrong default gateway (pointing to physical IP instead of virtual IP)

FIX: PC gateway must be the HSRP virtual IP (e.g., 192.168.10.1), not .2 or .3.

' ERROR: Port security violation — port shutdown

CAUSE: Third MAC address connected to port (hub, switch, or multiple devices)

FIX: "show port-security interface FaX/X" to see violation count. Clear: shutdown !' no shutdown.

' ERROR: show vlan brief shows no active ports

CAUSE: Ports assigned to VLAN before VLAN was created, OR port is in wrong mode

FIX: Create VLAN first, then assign. Verify port is "switchport mode access" not trunk.

' ERROR: VLANs not passing trunk

CAUSE: Allowed VLAN list on trunk does not include the VLAN

FIX: "switchport trunk allowed vlan add [vlan-id]" to add missing VLAN to trunk.

' ERROR: Switch will not accept "switchport trunk encapsulation dot1q"

CAUSE: 2960 switches only support dot1q — the command is not needed and will be rejected

FIX: Skip the encapsulation command. Just use "switchport mode trunk" on 2960 switches.

You did it.

From a blank Packet Tracer to a fully running enterprise network — in one day.

Technologies you can now confidently discuss in any CCNA interview:

- %_ VLANs & 802.1Q Trunking
- %_ Spanning Tree (Rapid PVST+)
- %_ OSPF Single Area
- %_ HSRP Gateway Redundancy
- %_ DHCP & Relay Agents
- %_ NAT / PAT Overload
- %_ Extended ACLs
- %_ SSH Hardening
- %_ DHCP Snooping & DAI
- %_ Port Security
- %_ Enterprise 3-Tier Design
- %_ Multi-Site WAN Architecture

Post your screenshots. Write your article. Tag recruiters.

You have earned this.